

Mobile Sales Analytics Dashboard - Portfolio Project

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Power BI Portfolio Case Study

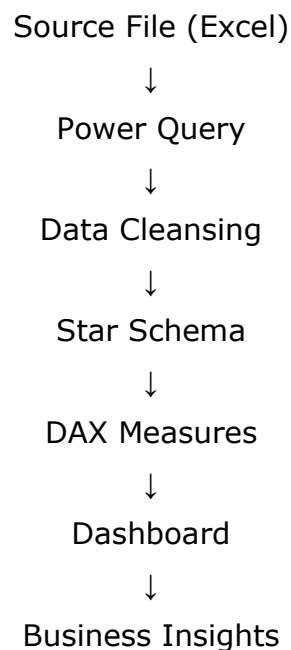
1. Executive Summary

This project demonstrates the design and development of an end-to-end Mobile Sales Analytics Dashboard using Power BI. The dashboard analyses 10,000+ mobile sales transactions and provides insights into sales performance, customer behaviour, brand performance, payment preferences, and geographic trends.

2. Business Problem

Management requires a centralized reporting solution to monitor mobile sales performance across cities, brands, customer demographics, and time periods. The objective is to transform raw transactional data into actionable insights that support data-driven decision making.

3. Project Architecture



4. Dataset Overview

Fields Used:

- Transaction ID
- Day, Month, Year
- Brand

- Mobile Model
- Price Per Unit
- Units Sold
- Customer Phone
- Customer Age
- City
- Payment Method
- Customer Ratings

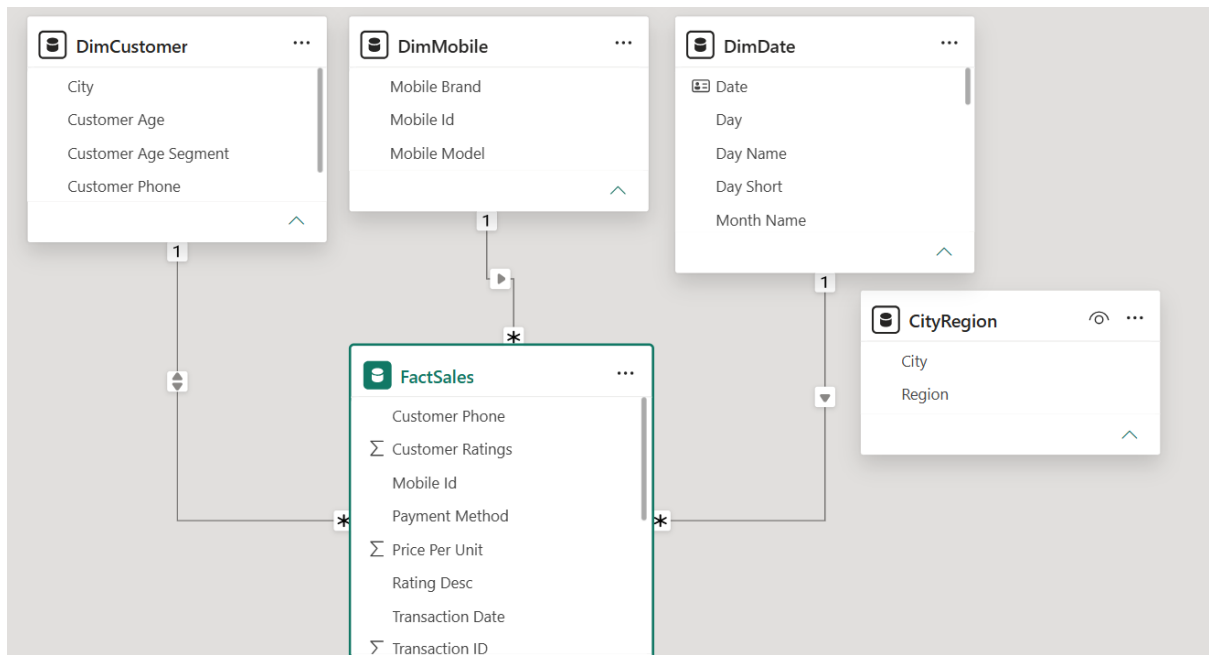
Dataset Statistics:

- Total Records: 10,000
- Mobile Brands: 5
- Mobile Models: 38
- Cities: 35
- Years Covered: 2021-2025
- Payment Methods: 4

5. Data Preparation & Modeling

- Performed data cleansing using Power Query
- Standardized text fields and data types
- Categorizing numeric Rating to Rating description
- Created Date Dimension table
- Applied Star Schema concepts
- Created Dimension tables for Products, Customers and City-Region mapping
- Established relationships between Fact and Dimension tables

STAR Schema:



6. DAX Measures

Key Measures:

- Total Sales

$$\text{Total Sales} = \text{SUMX}(\text{FactSales}, \text{FactSales}[\text{Units Sold}] * \text{FactSales}[\text{Price Per Unit}])$$

- Total Units Sold

$$\text{Total Quantity} = \text{SUM}(\text{FactSales}[\text{Units Sold}])$$

- Average Selling Price

$$\text{Average Price} = \text{AVERAGE}(\text{FactSales}[\text{Price Per Unit}])$$

- MTD Sales

$$\text{MTD} = \text{TOTALMTD}([\text{Total Sales}], \text{DimDate}[\text{Date}])$$

- QTD Sales

$$\text{QTD} = \text{TOTALQTD}([\text{Total Sales}], \text{DimDate}[\text{Date}])$$

- YTD Sales

$$\text{YTD} = \text{TOTALYTD}([\text{Total Sales}], \text{DimDate}[\text{Date}])$$

7. Dashboard Components

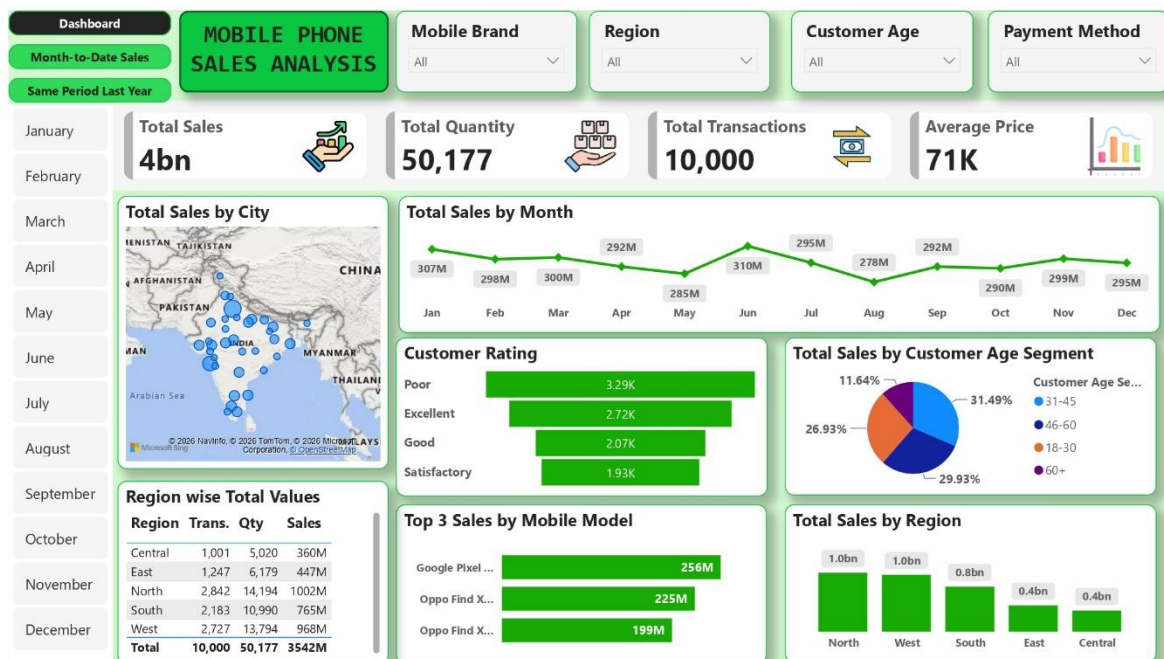
KPI Cards:

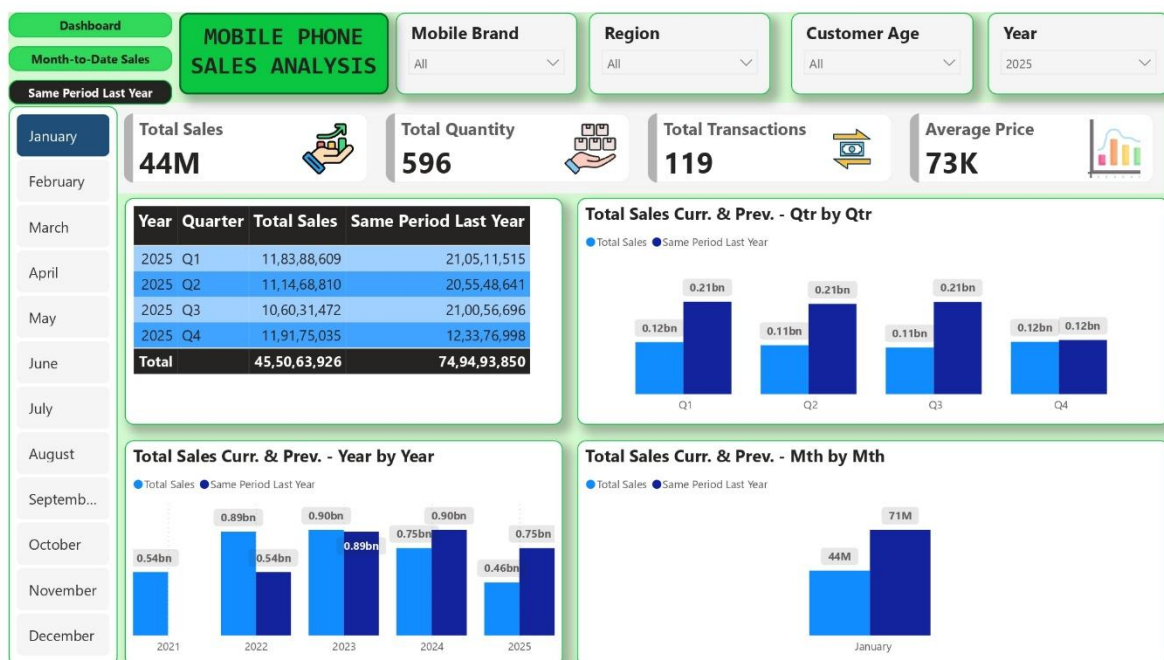
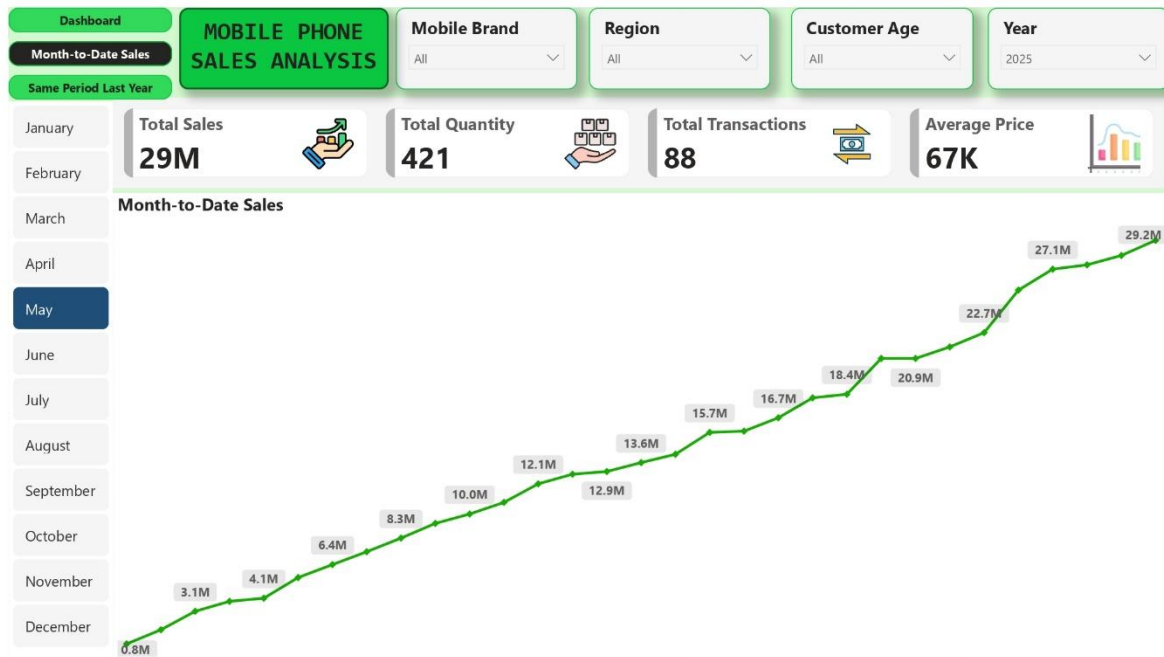
- Total Revenue
- Units Sold
- Average Selling Price
- Customer Rating

Visualizations:

- Monthly Sales Trend
- Brand-wise Revenue Analysis
- Model-wise Sales Analysis
- City-wise Sales Distribution
- Customer Age Segmentation
- Payment Method Analysis
- Customer Rating Distribution
- Top Selling Products

Page Layouts:





8. Key Business Insights

- Identified top-performing mobile brands and models.
- Analyzed customer purchasing behavior across age groups.
- Evaluated city-level sales performance.
- Identified preferred payment methods.
- Measured customer satisfaction through ratings analysis.
- Tracked sales performance over time using time-intelligence measures.

9. Key Findings

- Apple generated the highest revenue among all brands.
- North region contributed the highest share of total sales.
- Customers aged 31-45 accounted for the largest revenue segment.
- UPI was the most preferred payment method.
- Sales peaked during Q3 2025.

10. Business Value Delivered

- Enabled management to monitor sales performance across multiple dimensions.
- Reduced manual reporting effort through automated KPI tracking.
- Improved visibility into customer demographics and regional performance.
- Facilitated trend analysis through Time Intelligence measures.

11. Challenges & Solutions:

1. Denormalized Source Data

Solution: Created Fact and Dimension tables using Power Query.

2. Missing Region Information

Solution: Created custom City-Region mapping logic.

3. Time Intelligence Requirements

Solution: Built Date Dimension and implemented MTD/QTd/YTD measures.

12. Skills Demonstrated

- Power BI
 - Power Query
 - DAX
 - Data Modeling - Star Schema Design
 - Time Intelligence
 - Data Visualization
 - Dashboard Design
 - Business Analytics
 - KPI Development
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